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Clinicians-in-the-loop: A narrative review of clinician-facing integrated AI within mental health workflows

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STRUCTURED ABSTRACT

Objective:

This paper reviews clinician-in-the-loop (CITL) literature; AI tools, frameworks, and platforms to support mental health professionals in their workflows. We explored how CITL currently contributes to therapeutic skill development, decision-making, and clinical workflow integration. Implementation and integration strategies were also evaluated.

Materials and Methods:

We conducted a structured narrative synthesis to identify relevant empirical studies. Out of 984 articles, 13 papers met our inclusion criteria. Each selected paper described a clinician-facing AI system for therapy, training, or clinical decision support. We thematically categorized studies by primary function, design, and reported challenges.

Results:

A majority of the studies focused on early-stage prototypes demonstrating technical feasibility but limited real-world deployment. Common barriers included concerns about clinician autonomy, data privacy, and technological constraints. Efficacy depends not only on technical performance but also on alignment with real-world clinical practices. Clinicians preferred systems that enhanced their decision-making capabilities while maintaining their autonomy to assess and override algorithmic outputs.

Discussion:

We identified the following actionable priorities for ongoing CITL research efforts: (1) advancing from pilot to practice through iterative real-world validation; (2) reinforcing clinical skill development through embedded AI feedback and mentorship; (3) strengthening clinical decision support tools with contextualized human oversight; and (4) ensuring explainability, fairness, and accountability through alignment with clinical judgment and real-world constructs.

Conclusion:

CITL shows promise for improving and augmented mental health care, though most remain in early development. Future work should prioritize rigorous clinical evaluation, participatory design, and the operationalization of therapeutic constructs for valid, actionable feedback.

1. BACKGROUND AND SIGNIFICANCE

The increasing demand for mental health services and the simultaneous strain on the clinical workforce have accelerated interest in leveraging artificial intelligence (AI) to alleviate clinician burdens, including documentation, decision-making support, and therapeutic oversight. However, mental health care presents unique challenges. Therapeutic interactions require nuanced human judgment, empathy, and relational sensitivity – qualities that traditional AI systems can struggle to replicate.¹⁻⁴ Human oversight is recognized as essential for safely deploying AI in complex domains, where fully autonomous systems are often infeasible.^{5,6} Specifically for healthcare, there are serious challenges and concerns about the alignment of AI outputs with clinical norms and wellbeing needs, trust, safety, and overall care quality.^{7,8}

To address these challenges, the Human-in-the-Loop (HITL) framework^{5,6,9} has been explored in high-consequence domains to enable necessary safeguards and support transitional phases toward greater system autonomy. Recent research¹⁰⁻¹³ points to an adapted "Clinician-in-the-Loop" (CITL) model that positions clinicians within AI-assisted decision loops, ensuring oversight, interpretability, and human autonomy. For this paper, we define CITL as:

AI-enabled tools, frameworks, or platforms that embed mental health professionals within the process of reviewing, modifying, or overriding AI outputs at key decision points. These systems are designed to support clinical judgment and to promote patient-centered outcomes.

Despite growing recognition of CITL as central to integrating AI within high-stakes domains, its use in clinician-facing mental health workflows remains underexplored. A clear gap exists in understanding how CITL is designed, implemented, and evaluated in real-world mental health settings. This narrative review addresses this by synthesizing recent empirical studies of CITL, focusing on their integration into clinical workflows, impact on clinician decision-making and skill development, and implications for future development.

2. METHODS

2.1. Search Strategy

The narrative review focused on peer-reviewed, English-language empirical studies involving CITL. Eligible publications assessed applications of AI as an assistive tool in clinician workflow, decision-making, documentation, therapy augmentation, or trust development. Given the review scope, exclusion criteria were: other reviews, conceptual articles, study protocol publications, literature focusing only on patient-facing or researcher-facing uses, and model development.

We conducted a PubMed search in January 2025 to identify relevant papers. We used a Boolean search strategy by combining Medical Subject Headings (MeSH) and major topic

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(MAJR) terms, along with text keywords and phrases. The search strategy used OR operators to combine multiple related terms for each of the two primary concepts: clinical mental health and artificial intelligence. We then combined these two concepts using an AND operator. Iteratively refining the search strategies lead to using a NOT operator to exclude false positives. These methods ensured comprehensive and targeted search results by capturing all relevant permutations of terms while ignoring irrelevant topics.¹⁴ Table 1 details the used search terms.

We screened selected papers in two phases: first abstracts were examined for relevance, and then full-texts were evaluated. One author conducted the literature search, initial screening, and full-text assessment. A second author reviewed all included studies and corroborated final inclusion decisions. Given the focus of this review is not about exhaustive inclusion but interpretive breadth and contextual insight,¹⁵ we did not employ dual independent screening and inter-coder reliability statistics. Instead, any disagreements were resolved through reflexive deliberation, prioritizing relevance to the review’s conceptual scope.

With finalized inclusion articles, we used an interpretive and inductive process aligned with qualitative synthesis principles to review the full-texts and to define and identify themes. Articles were initially grouped by the primary goal of the clinician-facing tool, such as training or clinical decision support. This early sorting was guided by the authors’ interest in the systems’ function within clinical workflows and what specific actions they supported. We also extracted data about study design, model type, clinician role, and ethical considerations.

Thematic categories were refined iteratively by grouping concepts based on clinical intent, point of intervention within the clinician workflow, and the type of AI-clinician interaction facilitated. Though not guided by a formal codebook, the process resembled reciprocal translation, with studies interpreted in light of one another to identify conceptual commonalities and divergences.^{16–18} Thematic boundaries evolved in response to studies that did not cleanly fit initial categories, particularly those addressing implementation challenges and systemic barriers. This process drew on thematic synthesis strategies used in narrative reviews from adjacent healthcare fields,¹⁹ which evaluated mental health technologies under emergent practice conditions.

Table 1: Boolean search strategy with concept mapping applied to the PubMed database;
Concept 1 AND Concept 2 NOT Concept 3.

Concept 1: Clinical Mental Health	Concept 2: AI	Concept 3: Common Irrelevance
Cognitive Behavioral Therapy [MeSH]	Machine Learning [MeSH]	Surgical Procedures, Operative [MeSH]
	Artificial Intelligence [MeSH]	Remote Sensing Technology [MeSH Terms]
Counseling [MeSH]	Natural Language Processing [MeSH]	Smoking Cessation [MeSH Major Topic]
Crisis Intervention [MeSH]	Generative Artificial Intelligence	Locomotion [MeSH Terms]
Psychotherapy [MeSH]	Generative AI	
Mental Health Care	Generative Adversarial Network	
Psychotherapy/methods [MeSH]	Transformer-based architecture	
Professional-Patient Relations [MAJR]	Variational Autoencoder	
Suicide Prevention [MeSH]	Large Language Model	
Session Recordings	LLM	
Client Distress	Latent Space Model	
Conversations	Deep Generative Model	
Therapist-client interactions	Chatbot	
	Machine-Learning-Based Prediction	

3. RESULTS

Our initial search identified 984 potentially relevant articles. The first screening phase resulted in 20 articles, after removing duplicates, non-English, and ineligible articles given the exclusion criteria. Following the screening phase, eight articles^{20–27} met the inclusion criteria. An additional four articles, two peer-reviewed^{28,29} and two pre-print,^{30,31} meeting the inclusion criteria were identified following the citation networks of the initially selected eight articles and became available after the initial search. The resulting thirteen articles selected for inclusion were published between 2019–2025. Figure 1 illustrates the article selection process using the PRISMA³² framework. Supplemental Table 1 summarizes each article and denotes which of the four themes each was categorized under.

Nine studies focused on Cognitive Behavioral Therapy (CBT)^{28–31,33,34} or Motivational Interviewing (MI).^{28,35–37} Three studies^{29,31,38} evaluated general counseling skills training using AI-simulated clients, independent of specific therapeutic modality. Patient treatments included suicide prevention,³⁹ grief interventions,⁴⁰ ADHD behavioral therapy,²⁸ substance abuse disorder,³⁷ and PTSD.³⁰ Only one study³⁴ applied randomized controlled design; all others used feasibility testing, user evaluations, retrospective analyses, or qualitative/mixed-methods designs.

The qualitative thematic synthesis process resulted in four overarching themes. *Implementation Details* (n = 13) describes the technical design, deployment context, and integration of CITL into clinical workflows. *Tools for Skill Development and Assessment* (n = 10)^{21,23,25–31,35} refers to AI systems used to support clinician training, performance feedback, and therapeutic fidelity monitoring. *Clinical Decision Support Tools (CDSTs)* (n = 6)^{20,22,24–26,28} encompass systems that assist with real-time clinical decision-making, such as risk alerts or symptom monitoring. Three^{28,34,37} studies bridged the Skills Development and Assessment and the CDSTs domains by embedding clinician-facing dashboards that synthesized fidelity or adherence data into actionable insights, allowing clinicians to directly adjust care delivery based on AI-generated evaluations. *Challenges and Barriers* (n = 13) captured recurring issues related to the implementation, adoption, and effectiveness of CITL tools in real-world clinical settings.

3.1. Implementation Details

All 13 studies described technical design, deployment context, and clinician-facing CITL features. Key priorities included automated therapist feedback (n = 12),^{28–31,33–38,40,41} and real-time clinician interaction (n = 10),^{28–31,34,36–39,41} occasionally supported by large language models (n = 4).^{28–31} One study³⁸ examined a recurrent neural network. Table 2 summarizes each system's models, inputs, and outputs.

Eight studies^{20–22,24–26,28,35} used real therapy session data and five^{23,27,42–44} used simulated clinician-AI or clinician-scripted dialogues. These platforms typically employed automatic speech recognition (ASR), speaker diarization, behavior coding, and fidelity scoring pipelines to

analyze session audio. Six studies^{21,23,25,26,35,45} explored end-to-end systems that transcribed sessions, modeled therapist-client dynamics, and assigned feedback-driven outputs using frameworks like the MI Skill Code⁴⁶ or the Cognitive Therapy Rating Scale.⁴⁷ Seven studies^{21,23,25,26,35,42,45} used session audio as initial input, while four^{27,42-44} used session transcripts. Four studies^{20,22,24,26} used mixed input data, such as structured surveys, self-reports, behavior monitoring, or electronic health records. Audio pipelines were more common in fidelity monitoring tools; text inputs predominated in simulation-based training systems.

Five platforms^{23,29-31,38} enabled trainees to practice counseling through speech- or text-based interfaces. These tools provided structured, real-time feedback by labeling utterances, identifying strategies, and rating session quality. Seven systems^{22-25,43-45} incorporated dashboards to keep clinicians/trainees in the loop on their performance and/or patient care. One study³⁹ tailored implementation to a specific clinical context, i.e., embedding suicide risk alerts into a case management system for Native American youth.

3.2. Tools for Skill Development & Assessment

Ten studies^{21,23,25-31,35} focused on CITL as support for clinician skill acquisition and performance feedback. Four^{25,27,35,44} addressed general counseling techniques such as open-ended questioning and reflective listening, and four^{21,23,25,45} automated fidelity assessments to generate structured feedback.

Five simulation-based platforms^{23,27,42-44} enabled clinicians to practice therapeutic dialogue with AI-simulated patients using text- or speech-based interfaces. These systems provided granular, real-time feedback on counselor utterances, session structure, and strategy use. In contrast, systems using real-world audio/text^{20-22,24-26,28,35} typically offered session-level fidelity scores aligned with CBT or MI protocols.

Four studies^{21,23,25,35} provided structured fidelity feedback to generate session-level or turn-by-turn metrics; three^{27,42,43} used simulated interactions to give immediate or post-session feedback on specific counseling behaviors; and three^{26,44,45} delivered structured ratings, summaries, or dashboards to monitor performance trends. Six studies^{21,23,25,27,35,45} counted procedural behaviors (e.g., reflective statements, open questions, or affirmations) to output behavioral codes or summary scores. However, only two^{23,35} assessed the quality of these behaviors, particularly *Percent Complex Reflections*⁴⁶.

Three studies^{21,25,45} integrated mechanisms into supervision workflows, suggesting that CITL can support both individual skill refinement and structured clinical training models. One study⁴⁵ raised concerns about clinicians being "unsure how to respond to feedback" and did not "understand feedback metrics."

3.3. Tools for Clinical Decision Support

Six studies^{20,22,24–26,28} examined CITL implementations for assisting real-time clinical decision-making. These systems aimed to enhance clinician judgment by providing AI-generated alerts or recommendations derived from patient data. Use cases spanned risk prevention,²² symptom monitoring,^{20,24} dropout risk,²⁴ and live therapy guidance in various settings.^{25,26,42,45}

Most CDSTs were embedded into existing workflows, including electronic health records (EHR), and drew on structured data such as patient assessments or session-level information. For example, one²² applied predictive modeling to identify individuals exhibiting high suicide risk and generate case management alerts. Another, the *Trier Treatment Navigator*²⁴ tracked patient outcomes over time to support adaptive treatment planning (Lutz et al.). Tools like *Lyssn*⁴⁵ estimated dropout risk and visualized treatment progress using longitudinal EHR and session data. Meanwhile, *CBT-Trainer*⁴² delivered real-time, structured feedback during therapy delivery, helping clinicians adjust interventions mid-session based on protocol adherence.

3.4. Challenges & Barriers

All 13 studies identified barriers to CITL implementation and adoption, clustering around clinician acceptance, technical limitations, ethical concerns, and workflow integration. Three studies^{21,23,44} noted that their systems struggled to capture relational nuance (e.g., empathy, contextual sensitivity, and nonverbal behaviors) limiting reliability for assessing complex therapeutic skills. One²¹ reported that low feedback scores risked undermining clinician confidence. Privacy and consent concerns also arose,^{21,44} especially around session recording and data sharing. Two studies^{21,22} emphasized the need for transparent protocols to build trust among clinicians and patients.

Technical limitations included transcription and diarization errors that compromised behavior coding and fidelity scoring.³⁵ Clinicians in some studies^{43,45} found the feedback overly detailed, misaligned with priorities, or hard to interpret in-session. Others reported increased cognitive load and workflow disruption,^{21,25,35} along with challenges detecting rare behaviors and generalizing narrow fidelity metrics.^{23,27}

Workflow integration required changes to supervision practices and clinical infrastructure. Four studies^{21,25,43,45} described resource constraints related to session recording, data storage, and technical support. Clinicians needed additional training to interpret outputs and troubleshoot system issues,^{43,45} and in one case,²⁵ inconsistent access to assistance hindered sustained tool use. Other implementation challenges included lack of realism in AI simulations,²⁷ limited follow-up windows,^{26,42} platform access restrictions,⁴² and misalignment between clinical trial methods and AI evaluation frameworks due to small or imbalanced datasets.²⁰

Table 2: Structured comparison of the technical implementation components across the 13 included studies, including modeling methods, input modalities, and output modalities. Abbreviations: ‘ASR’, Automatic Speech Recognition; ‘LSTM’, Long short-term memory.

<u>Article</u>	<u>Quantitative Methods Used</u>	<u>Inputs</u>	<u>Outputs</u>
Brandl et al. ⁴⁰	Fuzzy cognitive map-based classification; Linear regression	Self-report questionnaires	Binary recommendation for offline support; explained variance in grief symptom change
Creed et al. ³³	NLP, signal processing analysis via <i>CORE-MI</i>	Session audio; cognitive therapy ratings	Session-level CBT fidelity scores; therapist feedback
Flemotomos et al. ³⁵	Clustering; ASR; <i>DeepSegment</i> ; Support vector regression; BiLSTM model	Session audio	Behavioral code predictions; MI session quality metrics; session client-patient dynamic metrics
Haroz et al. ³⁹	Machine learning suicide risk prediction model	Behavioral data; EHR; community data	Suicide risk alerts for case management
Imel et al. ³⁶	Support vector regression model; maxent language model; AST; NLP; signal processing	Session audio (simulated)	MI fidelity metrics; therapist performance feedback
Jeong et al. ²⁹	<i>ChatGPT-4o</i> ; mixed MANCOVA	Text utterances (simulated)	Structured AI-generated feedback, ratings on counseling skills
Lutz et al. ⁴¹	<i>LASSO</i> regression; nearest neighbor modeling; growth curve modeling	Patient characteristics & therapist session notes; session symptom tracking	Drop-out risks; adaptive treatment recommendations; real-time alerts
Peavy et al. ³⁷	ASR; machine learning classifier for MI coding	Session audio	Full transcripts; MI fidelity metrics; utterance predictions; counseling session visualizations

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<u>Article</u>	<u>Quantitative Methods Used</u>	<u>Inputs</u>	<u>Outputs</u>
Sadeh-Sharvit et al. ³⁴	ASR; NLP classification; automated note drafting	Session audio; patient self-report scales	Automated session summaries; CBT skill analysis; draft notes; aggregating supervision data; real-time feedback
Sibley et al. ²⁸	ASR; <i>Lyssn.io</i> ; transformer deep learning model	Session audio	MI fidelity scores & feedback; automatic <i>STAND</i> adherence scores; progress dashboards; performance badges
Stade et al. ³⁰	<i>TherapyTrainer</i> (LLM & prompt engineering to create an AI-patient and AI-consultant)	Text utterances (simulated)	Structured session and protocol-adherence feedback; protocol task guidance
Tanana et al. ³⁸	<i>ClientBot</i> (LSTM model); logistic regression classification	Text utterances (simulated)	Real-time basic skill classification & feedback (open questions, reflections)
Zhang et al. ³¹	LLM chatbot roleplays	Audio & text utterances (simulated)	Session logs of AI-simulated patient dialogues; AI feedback on CBT skills

4. DISCUSSION

The reviewed studies recurrently note a need for collaboration between clinical expertise and algorithmic support across various tasks, including documentation, triage, supervision, and therapeutic decision-making. Clinicians preferred AI systems that enhanced their decision-making capabilities while maintaining their autonomy to interrogate, contextualize, or override algorithmic outputs. We argue that CITL design can enable responsible integration of AI tools into mental health workflows. CITL focuses not only on technical performance but also on alignment with real-world clinical practices, with an emphasis on transparency, interpretability, and clinical autonomy preservation.

However, there is a lack of robust evidence supporting the efficacy and outcomes of the CITL approach. Our reviewed studies were mostly in early stages of development and lack long-term, real-world assessment. These constraints restrict the generalizability of current findings about how AI can impact real-world clinician decision-making, patient outcomes, and trust. To advance from isolated pilot projects to a robust evidence base, studies leveraging CITL must expand toward construct-valid, behaviorally grounded, and longitudinal assessments.

In the following subsections, we outline calls to action for future work in this domain: (1) advancing from pilot to practice through iterative real-world validation, (2) reinforcing clinical skill development through embedded AI feedback and mentorship, (3) strengthening clinical decision support tools with contextualized human oversight, and (4) ensuring explainability, fairness, and accountability by aligning AI outputs with clinical judgment and real-world practice. Each subsection integrates insights from recent conceptual frameworks and related fields to recommend future strategies for developing and deploying human-centered, clinically grounded AI systems. These actionable priorities respectively (a) propose construct-valid approaches to evaluation that emphasis the definition and measurement of intended clinical constructs,⁴⁸ and recommend (b) behavioral, longitudinal, and post-deployment metrics to assess real-world system impact.⁴⁹

4.1. Advance from pilot to practice through iterative real-world validation

CITL concepts must undergo rigorous, participatory validation in real-world settings.⁴⁸ They must also systematize core clinical concepts (e.g., therapeutic alliance support or risk alert efficacy) before operationalizing them into measurement tools.⁵⁰ Skipping this step risks mischaracterizing system effectiveness and missing critical safety concerns. Aligned with the field's broader push for measurement-based care, future deployments should explicitly define and validate targeted constructs to support clearer interpretation of outcomes.

Static benchmarks fall short in capturing the complexity of high-stakes mental health care.⁴⁹ Instead, evaluations should use longitudinal, context-sensitive methods that reflect patient diversity, clinician behavior, and ongoing human-AI interaction.^{51,52} Assessments should capture not only technical fidelity but also the effects on clinician decision-making and patient outcomes over time.⁵³ Real-world monitoring should also be treated as a continuous process;^{49,54} structured incident reporting, clinician feedback loops, and adaptive retraining mechanisms should be embedded in future implementations as they already are in other high consequence fields like aviation and transportation.^{55,56} Initial deployment should mark the beginning of an iterative cycle of measurement, refinement, and system realignment.

Achieving lasting impact requires evaluation standards and institutional infrastructure tailored to mental healthcare. While frameworks like CONSORT-AI improve transparency in other domains,⁵⁷ benchmarks suited to therapeutic context are lacking. The recently proposed READI framework directly addresses this gap, offering tailored evaluation criteria for assessing safety, privacy, equity, engagement, and implementation readiness in AI-mental health systems.⁵⁸ mental health lacks benchmarks suited to therapeutic context. WHO’s 2021 guidance emphasizes institutional governance for safe AI integration,⁵⁴ including clinical ethics committees,⁵⁹ clinician-led incident reporting,⁶⁰ audit-and-feedback loops,⁶¹ and regulatory oversight.^{62,63} The TEHAI framework reinforces this approach by assessing capability, utility, and ethical integration.⁶³ These conceptualizations offer a model for responsible AI deployment.

Evaluating impact also demands frameworks attuned to sociotechnical complexity. The non-adoption, abandonment, scale-up, spread, and sustainability (NASSS) framework,⁶⁴ links long-term adoption to organizational readiness, while implementation science and measurement-based care offer metrics for fidelity and therapeutic progress.^{65,66} Further, trust engineering frameworks can define measurable signals like calibrated reliance and contestability that future CITL iterations should monitor to preserve clinician–patient rapport.⁶⁷ Together, these frameworks can support responsive developments to clinical workflows and interpersonal dynamics.

4.2. Reinforce clinical skill development with embedded AI feedback and mentorship

Current AI-driven tools, while promising for enhancing clinician training, lack sufficient feedback granularity and contextual sensitivity. Most reviewed studies emphasized procedural skills (e.g., reflective statements, protocol adherence) while overlooking relational dynamics like empathy, alliance quality, or cultural responsiveness. This limits the broader utility of these systems, and reflects a conceptual gap with CITL: they often track whether behaviors occurred, not how skillfully they were delivered. Frequency counts may suffice for structured protocols, but offer little insight into therapeutic quality or clinician competence. Modeling behavior quality introduces new computational challenges, especially for systems prioritizing transparency, explainability, and contestability.⁶⁸

Structured definitions for such relational constructs are necessary before they can be meaningfully integrated within CITL applications.⁴⁸ While the reviewed tools monitor discrete actions, they fail to provide holistic essential feedback on clinician-patient interaction. Future systems should incorporate relational metrics to enable clinicians to assess and improve in these areas.⁶⁹ AI feedback that encourages reflection could bridge the gap between technical competence and clinical nuance.

Real-time feedback can be a core strength of future systems if it is iterative and also tailored to each clinician's learning curve and real-world complexity,^{49,70} Inspiration may be drawn from demonstrated in adjacent domains like surgical simulation,⁷¹ where micro-decision analysis and formative feedback have enabled tailored retraining for practicing clinicians. Future systems should also integrate continuous feedback loops that span multiple patient encounters,⁷² enabling clinicians to receive immediate input on procedural skills and longitudinal reflections on skill drift or development. This reflects recent conceptualizations of AI as a "teammate" that supports sustained skill acquisition and reflective learning across a clinician's career.⁷³

Mentorship will remain essential to contextualize AI feedback within broader supervision systems. Peer review, supervisory mentorship, and collaborative learning are currently irreplaceable for contextualizing AI insights and adapting to therapeutic nuance.^{5,74} Future systems should examine how to integrate AI feedback with mentorship to provide actionable, adaptive, and expedient insights that maintain accuracy and emotional responsivity to patient needs.

4.3. Strengthen CDSTs with contextualized human oversight

While the reviewed CDSTs show promise in enhancing decision-making through real-time data and personalized feedback, human oversight remains essential. Systems for suicide risk prediction and grief intervention provided timely, data-driven recommendations,^{39,40} but their success was relied on clinician engagement and seamless workflow integration. For real-world applicability, future implementations must undergo continuous clinical validation and adapt to both clinician and patient needs.⁴⁹

Real-time alerts and recommendations offer value, but must avoid compromising clinician autonomy and judgment. Clinicians should retain the ability to override or adapt AI suggestions based on each patient's unique context.⁴⁸ This balance is especially important for mental health, in which relational dynamics and individualized factors are central to care. Human oversight should remain a core feature, ensuring AI outputs are interpreted within the complexity of real-world therapeutic needs.⁷⁵ Maintaining agency is essential; CITL should support providers by offering contextual knowledge during selection between AI-generated options, rather than defaulting to rigid outputs.⁷⁶ As AI becomes more integrated into workflows, preventing clinicians from "falling out of the loop" will be a critical priority.

Among the reviewed tools, those with real-time feedback and session-level tracking like *CBT-Trainer* were better integrated into clinical practice.^{31,34} These systems delivered actionable feedback during sessions, helping clinicians make timely decisions. The effectiveness of these tools depended on their seamless integration into clinical workflows and clinician engagement. Systems that aligned with routine practices and offered timely, interpretable recommendations were better positioned to support decision-making without disrupting therapeutic rapport. However, their success hinged on clinician trust, which could erode if outputs felt overly prescriptive or misaligned with patient nuance. A balance between AI recommendations and clinical judgment remains vital, particularly in contexts that demand flexibility, empathy, and intuition.⁴⁹

Simulation-based platforms^{29–31,38} demonstrated that AI-supported feedback systems are effective for practicing therapeutic techniques, particularly when paired with mentorship. As in other high-skill fields like surgery and aviation, these tools can support clinician development.^{55,77–81} Integrating AI feedback into supervision models ensures that insights remain contextually grounded and clinically useful. Human mentorship helps ensure that AI augments rather than replaces expertise.

To maximize the effectiveness of CDSTs, future work should prioritize longitudinal evaluations of clinician and patient outcomes. Studies must go beyond technical benchmarks to examine the sociotechnical dynamics of human-AI interaction.⁴⁸ Embedding contextualized oversight into CDS tools will help preserve clinician autonomy, strengthen trust, and improve care outcomes.

4.4. Align AI outputs with clinical judgment and real-world constructs

AI systems must reflect the priorities of clinicians and the realities of therapeutic care. Clinicians’ skepticism about AI’s ability to capture nuanced elements is consistent across mental health literature,⁸² including a recent qualitative study examining mental health workers reactions to a hypothetical CITL treatment that monitors treatment fidelity.⁸³ These concerns underscore the need for AI outputs to be interpretable, context-aware, and integrated with collaborative workflows.^{3,69} Achieving this requires systematizing complex clinical concepts before embedding them into models to improve transparency and prevent misleading evaluations.⁴⁹

Effective integration also demands context-sensitive evaluation frameworks that reflect the complexity of therapeutic work. These must prioritize ethical principles and reinforce AI’s role as a support system for nuanced clinical reasoning and patient-centered care.⁸⁴ Real-world applicability requires dynamic, longitudinal evaluation rather than static benchmarks.⁴⁹ Continuous feedback loops should enable systems to evolve with clinician expertise and patient needs. Embedding structured human oversight can help balance AI recommendations with clinician autonomy and therapeutic nuance.⁸⁵

Trust also depends on transparent data practices that protect confidentiality, particularly when using session recordings or sensitive inputs. Several studies raised privacy and consent concerns; clear protocols are necessary to build confidence among clinicians and patients. Measuring impact through validated instruments is equally essential for system acceptance and accountability.^{48,86}

Future development should emphasize workflow compatibility and avoid overwhelming clinicians or diminishing their authority; real-time feedback must avoid disruptions to clinical decision-making. Iterative design, active clinician input, and post-deployment monitoring are key to ensuring that systems remain aligned with evolving care demands.^{49,87}

Achieving fairness and clinical relevance requires more than optimizing algorithmic performance. It also demands ethical grounding, stakeholder trust, and sensitivity to the contexts of care.^{88,89} Fairness must be defined through engagement with clinicians, patients, and institutions.^{90,91} Evaluation frameworks should extend beyond accuracy to include oversight structures, clinical trade-offs, and socially meaningful benchmarks.^{48,49} Establishing norms and institutions for ongoing evaluation is essential for embedding AI responsibly into clinical workflows. Combining trial-based rigor with adaptive monitoring can help center both system performance and human well-being.⁵⁷

4.5. Limitations

This review has several limitations. First, the early developmental stage and heterogeneity of included studies limited the effectiveness of indexed subject heading and keyword-based searches, and precluded a comprehensive statistical meta-analysis. Second, the narrative synthesis approach introduces potential interpretive bias due to subjective categorization and thematic grouping. Third, no self-assessment of bias was conducted during the initial screening of the 984 articles by a single reviewer. Further validation through longitudinal and prospective controlled studies is needed to support the conclusions drawn from current evidence. Lastly, while not strictly a limitation, two of the 13 included studies were unpublished pre-prints as of June 2025.

5. CONCLUSION

This review highlights the emerging role of clinician-in-the-loop (CITL) implementations to mental healthcare, synthesizing evidence from thirteen empirical studies of AI tools designed to augment clinical expertise. Unlike prior reviews that focus on patient-facing technologies, algorithmic accuracy, or generalized ethical considerations, this synthesis centers on clinician-facing tools embedded within real-world workflows. Through a structured narrative approach, we examined diverse systems that target therapeutic skill development, enhance clinical decision-making, and adapt to the complexities of clinical environments. Despite

technical innovation, most tools remain at early stages of development, with limited deployment and few evaluations focused on long-term integration, clinician behavior, or patient outcomes. Thematic analysis surfaced four recurring insights: heterogeneity in implementation strategies, frequent use of AI for skill training and fidelity feedback, targeted support for decision-making tasks, and persistent concerns around transparency, interpretability, and clinician autonomy. These findings underscore the need for rigorous, context-sensitive evaluation frameworks that reflect the relational, high-stakes nature of mental health care. To advance beyond proof-of-concept, CITL concepts should be validated in real-world practice and continuously refined through participatory feedback and performance monitoring. Systems should not only support decision-making but also scaffold therapeutic growth and preserve clinical judgment. With thoughtful design and oversight, CITL can enable more responsive, trustworthy, and equitable mental health care.

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COMPETING INTERESTS

The authors have no conflicts of interest nor competing interests to declare.

DATA AVAILABILITY

No datasets were generated or analyzed during the current study.

ETHICAL APPROVAL

Research with human or animal subjects was not conducted; therefore, ethical approval was not required.

LEGEND

Table 1: Boolean search strategy with concept mapping applied to the PubMed database; Concept 1 AND Concept 2 NOT Concept 3.

Table 2: Structured comparison of the technical implementation components across the 13 included studies, including modeling methods, input modalities, and output modalities. Abbreviations: ‘ASR’, Automatic Speech Recognition; ‘LSTM’, Long short-term memory.

Supplemental Table 1: Outlines of key information for each of the 13 included studies: citation details, brief description of the study’s goals and findings, and binary indicators for which of the four thematic categories each study addresses.

Figure 1: Flowchart illustration of the literature search and screening process. Includes counts of records identified, screened, excluded, and ultimately included in the review, following PRISMA guidelines.

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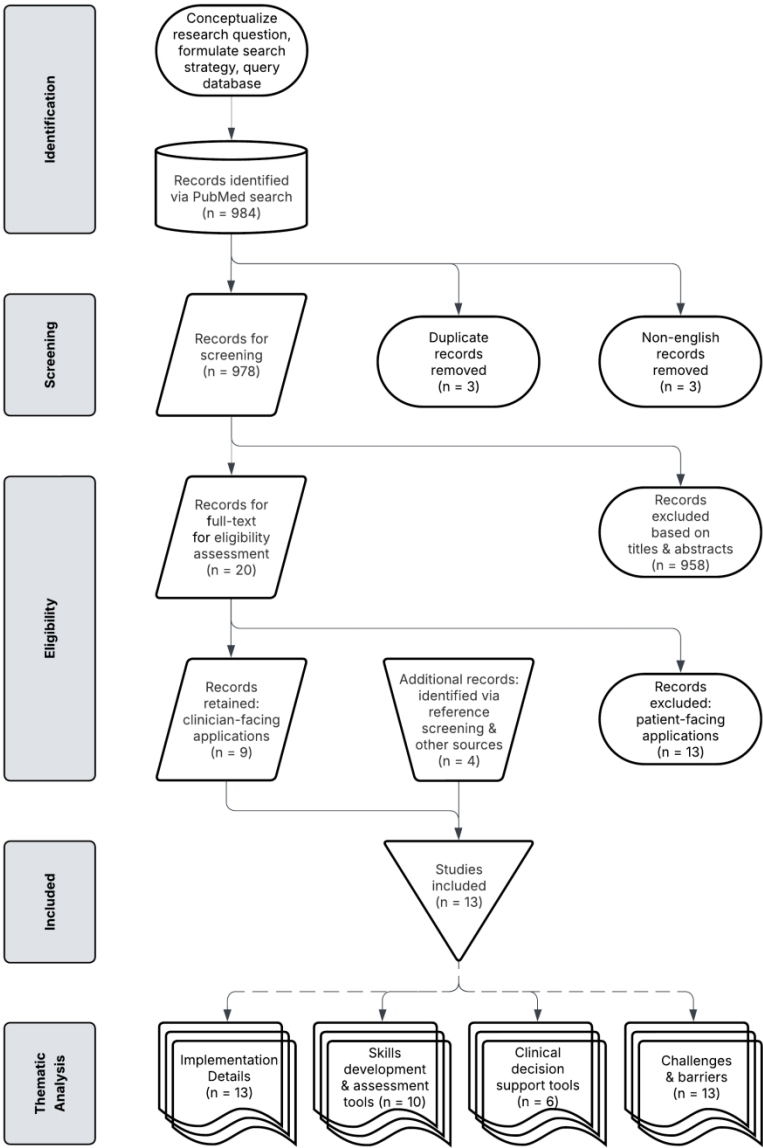


Figure 1: Flowchart illustration of the literature search and screening process. Includes counts of records identified, screened, excluded, and ultimately included in the review, following PRISMA guidelines.

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Supplemental Table 1: Outlines of key information for each of the 13 included studies: citation details, brief description of the study’s goals and findings, and binary indicators for which of the four thematic categories each study addresses.

<u>Article Title</u>	<u>Year</u>	<u>Authors</u>	<u>Themes</u>				<u>Short Summary</u>
			<u>Implementation</u>	<u>Skill Development</u>	<u>CDSTs</u>	<u>Challenges and Barriers</u>	
<i>Moving Toward Meaningful Evaluations of Monitoring in e-Mental Health Based on the Case of a Web-Based Grief Service for Older Mourners: Mixed Methods Study</i>	2024	Brandl et al. ⁴⁰	✓		✓	✓	An AI monitoring tool detected support needs in a grief intervention for older adults, achieving a high F1-score. Limitations involved skewed class distributions and challenges aligning clinical trial timing with AI evaluation.
<i>Knowledge and Attitudes Toward an Artificial Intelligence-Based Fidelity Measurement in Community Cognitive Behavioral Therapy Supervision³³</i>	2021	Creed et al. ³³	✓	✓		✓	Therapists evaluated demonstrations of an automated fidelity feedback tool (<i>CORE-CBT</i>). They valued its consistency but questioned relational accuracy. Limitations included small sample size and lack of follow-up tracking for real-world clinical outcomes.
<i>Automated evaluation of psychotherapy skills using speech and language technologies</i>	2021	Flemotomos et al. ³⁵	✓	✓		✓	An automated backend system provided therapist feedback through <i>CORE-MI</i> by transcribing, diarizing, and behaviorally coding sessions. Limitations included moderate transcription errors, misclassifications, and cascading mistakes across speech processing and behavioral coding modules.
<i>Designing a Clinical Decision Support Tool That Leverages Machine Learning for Suicide Risk Prediction: Development Study in Partnership With</i>	2021	Haroz et al. ³⁹	✓		✓	✓	A suicide risk prediction tool was adapted for Native American community workflows. Case managers used risk flags alongside clinical judgment. Limitations included small sample size, hypothetical deployment, and transparency and cultural fit concerns.

1								
2								
3	Native American Care							
4	Providers							
5								
6	Design Feasibility of an							An automated feedback system generated MI performance
7	Automated,							metrics and transcripts from standardized patient sessions.
8	Machine-Learning Based	2019	Imel et al. ³⁶	✓	✓		✓	Participants reported high usability and satisfaction.
9	Feedback System for							Limitations included reliance on simulated sessions and
10	Motivational Interviewing							variable agreement between machine and human coding.
11								
12								Counseling trainees used <i>ChatGPT</i> for client simulations and
13	AI Integration in Counseling							structured feedback. Participants improved self-efficacy and
14	Training: Aiding							reduced anxiety across sessions. Limitations included stronger
15	Counselors-in-Training in	2025	Jeong et al. ²⁹	✓	✓		✓	self-review effects and technical challenges maintaining
16	Self-Efficacy Enhancement							AI-client consistency during simulations.
17	and Anxiety Reduction							
18								
19								
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21								
22	Towards Integrating							Delivered machine learning-based feedback on drop-out risk
23	Personalized Feedback							and strategy selection to therapists. Limitations included
24	Research into Clinical	2019	Lutz et al. ⁴¹	✓		✓	✓	modest predictive power for some outcomes and reliance on
25	Practice: Development of the							retrospective development samples without prospective
26	Trier Treatment Navigator							validation.
27	(TTN)							
28								
29								<i>Lyssn.io</i> supported supervision in opioid treatment programs by
30	Improving the Quality of							recording sessions and delivering feedback. Focus groups
31	Counseling and Clinical							valued its educational potential but raised concerns about
32	Supervision in Opioid	2024	Peavy et al. ³⁷	✓	✓	✓	✓	workflow burden, patient consent, and cultural barriers to
33	Treatment Programs: How							recording and supervision adoption.
34	Can Technology Help?							
35								
36								
37	Effects of an Artificial							An AI platform provided session summaries, skill feedback,
38	Intelligence Platform for	2023	Sadeh-	✓	✓	✓	✓	and automated notes for therapists delivering CBT. Therapy
39	Behavioral Interventions on		Sharvit et					with AI support improved depression, anxiety, session
40	Depression and Anxiety		al. ³⁴					attendance, and documentation speed compared to
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Symptoms: Randomized Clinical Trial							treatment-as-usual, maintaining similar patient satisfaction.
Developed a community-focused implementation model for adolescent ADHD treatment (STAND) supported by AI technologies.	2025	Sibley et al. ²⁸	✓	✓	✓	✓	An AI platform supported community therapists delivering adolescent ADHD treatment. Therapists received feedback on MI skills and protocol adherence. Pilot results showed feasibility but raised barriers around supervisor engagement and trust in AI evaluations.
TherapyTrainer: Using AI to Train Therapists in Written Exposure Therapy	2024	Stade et al. ³⁰	✓	✓		✓	An AI training tool allowed therapists to practice Written Exposure Therapy with AI-simulated patients. Participants reported increased preparedness and confidence. Challenges included feedback overload, usability barriers, and session guidance misalignment with therapist workflow.
Development and Evaluation of ClientBot: Patient-Like Conversational Agent to Train Basic Counseling Skills	2019	Tanana et al. ³⁸	✓	✓		✓	A machine learning conversational agent simulated therapy sessions and delivered immediate feedback on open questions and reflections. Participants improved skills but noted lower engagement compared to human partners during training sessions..
CBT Trainer: AI Patient for Training CBT Therapists — a Pilot Study	2025	Zhang et al. ³¹	✓	✓		✓	A training platform used AI-simulated patients and structured feedback to train therapy trainees. Participants improved in assessment and information gathering skills. Challenges included variability in trainee engagement and feedback preferences across sessions